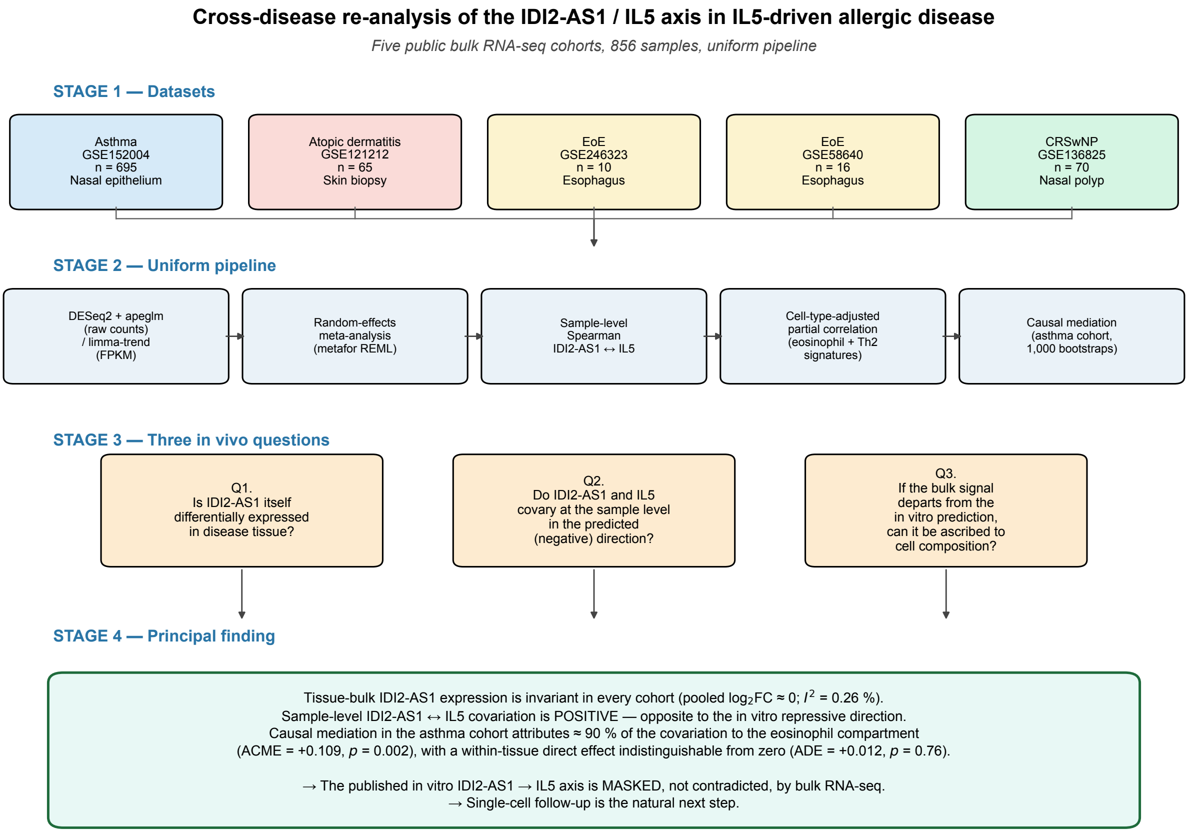
Figure 1


Cross-disease re-analysis of the IDI2-AS1 / IL5 axis in IL5-driven allergic disease
Five public bulk RNA-seq cohorts, 856 samples, uniform pipeline
STAGE 1 — Datasets
Asthma GSE152004 n = 695 Nasal epithelium
Atopic dermatitis GSE121212 n = 65 Skin biopsy
EoE GSE246323 n = 10 Esophagus
EoE GSE58640 n = 16 Esophagus
CRSwNP GSE136825 n = 70 Nasal polyp
STAGE 2 — Uniform pipeline
DESeq2 + apeglm (raw counts) / limma-trend (FPKM)
Random-effects meta-analysis (metafor REML)
Sample-level Spearman IDI2-AS1 ↔ IL5
Cell-type-adjusted partial correlation (eosinophil + Th2 signatures)
Causal mediation (asthma cohort, 1,000 bootstraps)
STAGE 3 — Three in vivo questions
Q1. Is IDI2-AS1 itself differentially expressed in disease tissue?
Q2. Do IDI2-AS1 and IL5 covary at the sample level in the predicted (negative) direction?
Q3. If the bulk signal departs from the in vitro prediction, can it be ascribed to cell composition?
STAGE 4 — Principal finding
Tissue-bulk IDI2-AS1 expression is invariant in every cohort (pooled $\log_2$FC ≈ 0; I^2 = 0.26 %).
Sample-level IDI2-AS1 ↔ IL5 covariation is POSITIVE — opposite to the in vitro repressive direction.
Causal mediation in the asthma cohort attributes ≈ 90 % of the covariation to the eosinophil compartment
(ACME = +0.109, p = 0.002), with a within-tissue direct effect indistinguishable from zero (ADE = +0.012, p = 0.76).
→ The published in vitro IDI2-AS1 → IL5 axis is MASKED, not contradicted, by bulk RNA-seq.
→ Single-cell follow-up is the natural next step.